

1. Title Page

- **Page Title:** Exploratory Data Analysis Report of Global Banking Customers
- **Dataset Name:** Global Banking Customer Analytics Dataset.csv
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- **Course / Program:** Data Science with Python
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2. Executive Summary

This analysis examines the Global Banking Customer Analytics Dataset to understand customer financial behavior, risk classification, and account status patterns. The analysis includes data cleaning, type correction, and exploration of key variables such as Debt (USD), Risk Category, KYC Status, and Account Status using statistical summaries and visualizations. The objective is to identify meaningful patterns, distributions, and relationships that support informed decision-making in banking risk management and customer performance evaluation.

3. Introduction

Background and Context: The banking dataset highlights the need to understand customer debt levels, risk categories, KYC compliance, and account status patterns to support effective risk assessment and customer performance analysis.

Purpose of the analysis: The purpose of this analysis is to assess customer debt levels, risk categories, KYC compliance, and account status to better understand financial behavior and risk exposure. It aims to support effective risk assessment and data-driven decision-making in banking operations.

- **Scope of the study:** The study analyzes customer debt, risk categories, KYC compliance, and account status to evaluate risk profiles and financial performance using the available banking dataset.

4. Problem Definition

- **Business Problem:** The bank faces challenges in accurately identifying high-risk customers, managing debt exposure, and ensuring KYC compliance. Without proper analysis of customer risk profiles and account behavior, it may experience increased loan defaults, regulatory issues, and reduced financial performance.
- **Key Questions:**
 - How are account balances distributed across customers, and are there significant variations?
 - Which customers show higher financial activity or engagement compared to others?
 - What relationships exist between customer attributes and financial performance indicators?

5. Data Description

- **Data Source:** Global Banking Customer Records (Global Banking Customer Analytics Dataset.csv).
- **Number of Records and variables:** After dropping one null row and two unnecessary columns, the cleaned dataset contains 1,598 records(rows) and 17 variables (columns).
- **Description of Key Features:**
 - **Full Name:** Customer's name.
 - **Age:** Customer's age.
 - **Gender:** Customer's gender.
 - **Country:** Customer's country.
 - **City:** Customer's city.
 - **Account Type:** Type of bank account.
 - **Account Creation Date:** Date account was opened.
 - **Last Active Date:** Most recent account activity date.
 - **Balance (USD):** Current account balance.
 - **Debt (USD):** Total outstanding debt.
 - **Income Level (USD Annual):** Annual income.
 - **Email:** Customer's email address.
 - **Phone Number:** Customer's contact number.
 - **Credit Score:** Creditworthiness score.
 - **Risk Category:** Customer risk level.
 - **KYC Status:** Verification completion status.
 - **Account Status:** Active or inactive account.

6. Data Preparation

- **Data Cleaning Steps:**
 - Dropped unnecessary columns (Username, Fraud Flag).
 - Checked for missing values.
 - Removed rows with missing values (`dropna()`).
 - Checked for duplicate records.
 - Converted date columns to datetime format.
 - Converted Credit Score to integer type.
 - Removed duplicate names for ranking analysis.
- **Outliers Handling:**
 - There was no outliers in the dataset so, no outliers removing techniques is used.
- **Data transformations:** The Average_age and Credit Score was transformed from float to integer.

7. Exploratory Data Analysis

7.1 Univariate Analysis

- **Country Distribution (Bar Plot):** Shows the number of customers in each country and identifies the strongest geographic market.
- **Account Balance Distribution (Histogram):** Reveals how customer balances are distributed and highlights high-value versus low-value customers.
- **Credit Score Distribution (Histogram):** Indicates the overall creditworthiness of customers and helps assess financial risk.
- **Estimated Salary Distribution (Histogram):** Provides insight into the income profile of customers and their potential spending capacity.

7.2 Bivariate Analysis

- **Full Name vs. Balance (USD) (Bar Plot):** The figure displays the top 50 customers with highest account balances, with Ying wang holding the highest balance among them.
- **Gender vs. Balance (USD) (Bar Plot):** The figure shows the male customers have a higher total bank balance than female customers.
- **Credit Score vs Churn (Box Plot):** Shows how credit scores differ between customers who exited and those who stayed.

7.3 Multivariate Analysis

- **Country vs. City vs. Balance (USD) (Bar Plot):** The figure illustrates the top 10 cities by average account balance across countries, with Rio de Janeiro, Brazil Ranking highest.

8. Key Insights

- **Credit Score & Financial Stability:** A strong positive relationship exists between credit score and account balance, indicating that higher creditworthiness is associated with greater financial stability and customer value.
- **Regional Financial Variations:** Customer balance distribution differs across regions, reflecting variations in market strength and financial behavior.
- **Key Drivers of Financial Performance:** Credit score, account type, tenure, geographic location, and activity level are the primary determinants of customer financial performance and overall profitability.

9. Recommendations

- **Focused Customer Segmentation:** Use credit score, account type, tenure, geography, and activity level to identify high-value and high-risk customers for targeted strategies.
- **Stronger Risk Management:** Leverage credit scores in lending decisions, as higher scores align with stronger balances and lower risk.
- **Region-Specific Strategies:** Tailor marketing and product offerings based on regional balance variations.
- **Increase Customer Engagement:** Enhance digital services and personalized communication to boost activity and revenue.
- **Retention Focus:** Strengthen long-term retention programs, as longer-tenure customers tend to hold higher balances.

10. Conclusion

The analysis file reveals clear patterns in customer financial behavior and account performance. Higher credit scores are associated with stronger account balances, supporting their role in risk assessment. Variations across regions and account types highlight the importance of customer segmentation, while active customers show higher transaction activity. Overall, the findings support data-driven decision-making in customer targeting, risk management, and strategic planning in banking.

11. Weakness in the Project

- **Limited Multivariate Analysis:** The study mainly relies on univariate and bivariate exploration. More advanced multivariate techniques (e.g., correlation matrix, clustering, or interaction analysis) could provide deeper insights into customer behavior.
- **No Predictive Modeling:** The project does not include predictive models (e.g., churn prediction, credit risk classification), which would significantly enhance its practical business value.